Giacomo Fenzi

Personal Information

Email:	giacomo.fenzi@epfl.ch
Website:	`https://gfenzi.io`
Profiles:	DBLP, Google Scholar, GitHub
Birth place & date:	Milan (Italy), 28 March 1998
Nationality:	Italian
Languages:	English (fluent), Italian (native)

Research Interests

Cryptography, theoretical computer science, proof systems, zkSNARKs, IOPPs, Reed–Solomon codes, lattices, and sumcheck

Academic experience

- **EPFL, Ph.D.**, Computer Science, September 2022 – present (expected to graduate September 2026)
  Thesis: *Query-Optimal Interactive Oracle Proofs from Code Switching*
  Advised by Alessandro Chiesa
  Attended the Simons Institute *Cryptography 10 Years Later: Obfuscation, Proof Systems, and Secure Computation* program in May to August 2025

- **ETH Zürich, M.Sc.**, Cybersecurity, September 2022
  Thesis: *Klondike: Finding Gold in SIKE*
  Advised by Kenneth Paterson and Fernando Virdia

- **University of St. Andrews, B.Sc.**, Computer Science and Mathematics, July 2020
  Thesis: *Zero Knowledge Proofs Theory and Applications*
  Advised by Stephen Linton and Louis Theran

Awards and Grants

- Best Paper Award at CRYPTO 2024 (recognizes the paper "STIR: Reed–Solomon Proximity Testing with Fewer Queries" for a breakthrough result)

- Teaching Assistant Award 2024 at EPFL (awarded for exceptional contribution to teaching excellence)

- Ethereum Foundation Academic Grant (awarded \$80'000 for *Post-quantum signatures for Ethereum from hash-based succinct arguments*).

- Academic Research Program Grant (awarded \$25'000 for *Small and efficient multilinear polynomial commitments in the pure ROM*).

Publications

2026

1. *TensorSwitch: Nearly Optimal Polynomial Commitments from Tensor Codes*
   Benedikt Bünz, Giacomo Fenzi, Ron D. Rothblum, William Wang
   CRYPTO 2026 and **ZKProof 8**

2. *Finding Bugs and Features Using Cryptographically-Informed Functional Testing*
   Giacomo Fenzi, Jan Gilcher, Fernando Virdia
   RWC 2026 and **TCHES 2026**

3. *Interactive Proofs for Batch Polynomial Evaluation*
   Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, Eylon Yogev
   ITC 2026

4. *Open Problems in List Decoding and Correlated Agreement*
   Gal Arnon, Dan Boneh, Giacomo Fenzi
   Cryptology ePrint Archive, Paper 2026/680

5. *Zero-Knowledge IOPPs for Constrained Interleaved Codes*
   Alessandro Chiesa, Giacomo Fenzi, Guy Weissenberg
   ZKProof 8

6. *Field-Agnostic SNARKs with Small Proofs via Encode-Repeat-Accumulate (ERA) Codes*
   Anubhav Baweja, Giacomo Fenzi, Pratyush Mishra, Tushar Mopuri
   ASIACRYPT 2026

7. *How to Prove More False Statements: Fiat–Shamir Limitations on (Generated) R1CS*
   Giacomo Fenzi
   Cryptology ePrint Archive, Paper 2026/1838

2025

1. *WHIR: Reed–Solomon Proximity Testing with Super-Fast Verification*
   Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, Eylon Yogev
   EUROCRYPT 2025 and **zkSummit 12**

2. *Linear Time Accumulation Schemes*
   Benedikt Bünz, Alessandro Chiesa, Giacomo Fenzi, William Wang
   TCC 2025, **CAW 2025**, **zkSummit 13**, and **ZKProof 8**

3. *Time-Space Trade-offs for the Sumcheck Protocol*
   Anubhav Baweja, Alessandro Chiesa, Elisabetta Fedele, Giacomo Fenzi, Pratyush Mishra, Tushar Mopuri, Andrew Zitek-Estrada
   TCC 2025 and **ZKProof 8**

4. *A Time-Space Tradeoff for the Sumcheck Prover*
   Alessandro Chiesa, Elisabetta Fedele, Giacomo Fenzi, Andrew Zitek-Estrada

TCC 2025 and **ZKProof 6**

5. *Post-Quantum Access Control with Application to Secure Data Retrieval*
   Behzad Abdolmaleki, Hannes Blümel, Tianxiang Dai, Giacomo Fenzi, Homa Khajeh, Stefan Köpsell, Maryam Zarezadeh
   Communications in Cryptology 2025

6. *Small-field Hash-based SNARGs Are Less Sound than Conjectured*
   Giacomo Fenzi, Antonio Sanso
   Cryptology ePrint Archive, Paper 2025/2197

7. *zip: Reducing Proof Sizes for Hash-Based SNARGs*
   Giacomo Fenzi, Yuwen Zhang
   Cryptology ePrint Archive, Paper 2025/1446

2024

1. *STIR: Reed–Solomon Proximity Testing with Fewer Queries*
   Gal Arnon, Alessandro Chiesa, Giacomo Fenzi, Eylon Yogev
   CRYPTO 2024, **zkSummit 11**, and **ZKProof 6**
   Best Paper Award (awarded to two papers at the conference)

2. *zkSNARKs in the ROM with Unconditional UC-Security*
   Alessandro Chiesa, Giacomo Fenzi
   TCC 2024 and **ZKProof 6**

3. *Lova: Lattice-Based Folding Scheme from Unstructured Lattices*
   Giacomo Fenzi, Christian Knabenhans, Khanh Ngoc Nguyen, Duc Tu Pham
   ASIACRYPT 2024

4. *SLAP: Succinct Lattice-Based Polynomial Commitments from Standard Assumptions*
   Martin R. Albrecht, Giacomo Fenzi, Oleksandra Lapiha, Ngoc Khanh Nguyen
   EUROCRYPT 2024 and **ZKProof 6**

5. *Lattice-Based Polynomial Commitments: Towards Concrete and Asymptotical Efficiency*
   Giacomo Fenzi, Hossein Moghaddas, Ngoc Khanh Nguyen
   Journal of Cryptology 2024 and **ArcticCrypt 2025**

Books

1. *Latin Diachronic Frequency Dictionary Vol. 2.*
   Giacomo Fenzi.
   Propylaeum: Digital Classics Books, Universitätsbibliothek Heidelberg, 2022.

2. *Latin Diachronic Frequency Dictionary Vol. 4.*
   Giacomo Fenzi, James Leslie, William Short, Thomas Spinelli.
   Propylaeum: Digital Classics Books, Universitätsbibliothek Heidelberg, 2022.

Academic Service

– Judge, The Proximity Prize, a $1M prize for research on long-standing coding-theory conjectures, and the autoresearch competition hosted at better.codes

– Co-organizer, Ethereum Post-Quantum Workshop, 10–11 October 2026

– Organizer, IOPFest, 9 May 2026

Work Experience

– **Succinct Labs**, San Francisco, United States
Research Intern, May 2025 – September 2025

– **ETH Zürich**, Zürich, Switzerland
Hilfsassistent, September 2021 – August 2022

– **Twitter**, London, United Kingdom
Intern, Make Everything Searchable, June 2021 – September 2021

– **University of St Andrews**, St Andrews, United Kingdom
Research Assistant, June 2020 – August 2020

– **Goldman Sachs**, London, United Kingdom
Intern, Merchant Banking Division, June 2019 – August 2019

– **Deloitte**, Milan, Italy
Intern, Advanced Analytics, June 2018 – July 2018

– **TecGlass Digital**, Lalín, Spain
Intern, June 2017

Talks and Invited Lectures

Interactive Proofs for Batch Polynomial Evaluation

– 2026–08–15/16, ITC 2026 (Santa Barbara, CA, USA; scheduled)

Polylogarithmic Proofs for Multilinears over Binary Towers

– 2026–05–14, EUROCRYPT 2026 (Rome, Italy)

Zero-Knowledge IOPPs for Constrained Interleaved Codes (ZO0K)

– 2026–05–09, ZKProof 8 (Rome, Italy)

– 2026–05–06, zkSummit (Rome, Italy)

WHIR: Reed–Solomon Proximity Testing with Super-Fast Verification

– 2026–01–20, Starkware (online)

– 2026–01–14, Aztec (online)

– 2025–05–04/08, EUROCRYPT 2025 (Madrid, Spain)

– 2025–03–31, Security and Privacy Lab (University of Pennsylvania, online)

– 2025–03–23/25, ZKProof 7 (Sofia, Bulgaria)

– 2025–02–18, Real World Crypto (Paris, France)

– 2024–11–28, IBM Research (Zurich, Switzerland)

– 2024–11–08, Nexus Speaker Series and Sumcheck Builder Group (online)

– 2024–10–30, BSA EPFL (Lausanne, Switzerland)

Small-field Hash-based SNARGs Are Less Sound than Conjectured

– 2025–12–19, Ethproofs (online)

The Proximity Prize

– 2025–11–07, Ethproofs (online)

Linear Time Accumulation Schemes (WARP)

– 2025–10–31, Swiss Crypto Day (Lausanne, Switzerland)

– 2025–06–02, Stanford University (Stanford, CA, USA)

– 2025–05–04, CAW (Madrid, Spain)

Hash-Based Succinct Arguments

– 2025–08–13/14, Google ZK & AI Summit (Sunnyvale, CA, USA)

A Survey on Efficient Succinct Arguments

– 2025–06–20, Cryptography: 10 Years Later (Simons Institute, Berkeley, CA, USA)

STIR: Reed–Solomon Proximity Testing with Fewer Queries

– 2024–10–30, BSA EPFL (Lausanne, Switzerland)

– 2024–10–16, Nexus Speaker Series and Sumcheck Builder Group (online)

– 2024–09–12/13, TUM Blockchain Conference (Munich, Germany)

– 2024–09–02, Swiss Crypto Day (St Gallen, Switzerland)

– 2024–08–18/22, CRYPTO 2024 (Santa Barbara, CA, USA)

– 2024–07–03, Starkitecture (online)

– 2024–05–22/24, ZKProof 6 (Berlin, Germany)

– 2024–04–08, zkSummit 11 (Athens, Greece)

UC-Secure zkSNARKs

– 2024–12–02/06, TCC 2024 (Milan, Italy)

– 2024–10–30, BSA EPFL (Lausanne, Switzerland)

– 2024–05–22/24, ZKProof 6 (Berlin, Germany)

– 2024–03–11, NYU (New York, NY, USA)

– 2024–03–08, University of Pennsylvania (Philadelphia, PA, USA)

– 2024–03–04, BUSec (Boston, MA, USA)

Lova: Lattice-Based Folding Scheme from Unstructured Lattices

– 2025–03–17, ENSL/CWI/KCL/IRISA Joint Cryptography Seminars (online)

SLAP: Succinct Lattice-Based Polynomial Commitments from Standard Assumptions

– 2024–05–26/30, EUROCRYPT 2024 (Zurich, Switzerland)

– 2024–05–22/24, ZKProof 6 (Berlin, Germany)

– 2023–12–15, CWI (Amsterdam, Netherlands)

Lattice-Based Polynomial Commitments: Towards Concrete and Asymptotical Efficiency

– 2025–07–06/11, ArcticCrypt 2025 (Longyearbyen, Svalbard)